What version 4 does

Version 0.4 organizes evidence and generates **retrospective research reports** on real wildfires. You can work with the ten prefilled cases or explore the more detailed Gironde pilot in France.

Workflow	Inputs	Outputs
Ten documentary cases	30 prefilled records: 3 per case.	One report per run, with findings, sources, gaps and next steps.
Gironde pilot	3 original Copernicus PDF maps; 6 extracted records.	A report showing which products and records are admitted at each temporal cutoff under the adopted bounds.

Results appear as text in the **Analysis** tab and can be exported as research files in a ZIP archive. The system does not generate new satellite imagery, fire-spread forecasts or operational evacuation maps.

Start with this test, without Gemini

1. Extract the entire v0.4 ZIP and run **main.py**.
2. In **Reviewed pilot**, click **Open first reviewed dossier**.
3. The application switches to **Analysis**, where a prepared report is ready to read.
4. Return to **Reviewed pilot**, keep **PRIMARY** selected and click **Run pilot cutoff** to create a new run.

The first button opens the prepared report. The second runs the analysis again and saves a new result. Both work without Gemini when it is disabled.

How to use this manual

Names in **bold** match the application controls. If a tab is hidden, use the **Open view** selector at the bottom of the window. These instructions were checked against the v0.4 source code; the Windows appearance may vary.

Set up and launch on Windows

1. Extract the complete package

Use **FireWatch_AI_Real_Cases_v0_4_Final.zip** or the complete **FireWatch_AI_Real_Cases_GitHub_Ready.zip** package. Right-click the ZIP and select **Extract All**. Open the extracted folder containing main.py.

Keep the **data**, **originals** and **pilot** folders together with the other project files. Do not run main.py inside the ZIP or copy only that file.

2. Launch through PyCharm

1. In PyCharm, select **File → Open** and choose the folder containing main.py.
2. Check **Settings → Project → Python Interpreter** and select Python 3.12. The exact menu name may vary with your installation.
3. In the project tree, right-click **main.py**.
4. Select **Run 'main'**.

The application uses the Python standard library and Tkinter; no pip packages are required. The window title should include **Final research release v0.4**.

Alternative: launch without PyCharm

Double-click **run_gui.bat** in the extracted folder. It looks for Python 3.12 using **py -3.12**. If it fails, run main.py through PyCharm with the correct interpreter.

3. Recognize the starting screen

Version 0.4 opens with **FW-RC-008 / Gironde** selected and the **Reviewed pilot** tab visible. The top selector changes the case; the bottom **Open view** selector changes the tab.

For your first test, leave **Enable Gemini for analysis** unchecked in **Gemini settings**. Offline analysis works with the included data.

Where your data is saved

Runs and reviews are saved automatically in **workspace/firewatch.sqlite3** inside the project folder. Keep this folder to preserve your history. A ZIP containing results is saved only when you request an export.

Run the Gironde pilot

A. Read the prepared report

Open **Reviewed pilot → Open first reviewed dossier**. The application selects Gironde, loads run **FW-RC-008-PILOT-BASELINE-v1** and opens **Analysis**. This button does not make a new API request.

B. Check the source PDFs

In **Reviewed pilot**, use the product selector to choose P01, P02 or P03, then click **Open original PDF**. The map opens in your computer's PDF reader. Check its map information, legend and sources. Repeat for the other products.

C. Generate a new report

1. In **Reviewed pilot**, select a temporal cutoff.
2. Keep **PRIMARY** for your first test.
3. Click **Run pilot cutoff** and wait for completion.
4. Read the result in **Analysis**. The bottom status bar should show **Saved run**.
5. To compare results, return to the pilot tab and repeat with EARLY and LATER.

Cutoff (UTC)	Expected offline result
EARLY · 2022-07-17 00:00	No products admitted; none of the 6 records is included. This is a valid cutoff result.
PRIMARY · 2022-07-17 12:00	Only P01, the first Landiras map: 2 records included; 4 excluded.
LATER · 2022-07-18 06:00	P01, P02 and P03: all 6 records included.

What is a cutoff? It is the reference time used to assess the evidence. The times above are in UTC, as in the interface. Keep these UTC values when reproducing this test.

Why “conditional”? The portal delivery timestamp has no timezone, and some metadata remains unresolved. The system uses conservative intervals. Admitting a product does not prove that response teams received it at that time.

For PRIMARY, the expected offline engine is **deterministic_bounded_source_review** and the status is **CONDITIONAL_SOURCE_REVIEW_COMPLETED**. The included technical review was performed by AI; human review remains pending until you record it.

Run all ten cases and read results

One case at a time

1. Choose a case in the top selector.
2. Open **Prefilled evidence** to inspect its three records, sources and locators.
3. Open **Analysis** and select **Retrospective documents (prefilled)**.
4. Click **Run analysis**. Do not enter a cutoff in this mode.
5. Wait for **Saved run** and read the report in Analysis.

All ten cases in one batch

1. In **Analysis**, keep **Retrospective documents (prefilled)** selected.
2. Click **Analyze all 10 cases**.
3. Wait for **10/10 document analyses completed**. The central panel displays a batch summary.
4. To read a full report, select its case at the top.
5. Open **History & review** and choose a run under **Saved run**.
6. Return to **Analysis** to read the text. Selecting a saved run loads the report but does not automatically switch tabs.

The batch uses the 30 documentary records across the ten cases. It does not automatically execute the three pilot cutoffs. Use Reviewed pilot separately to compare EARLY, PRIMARY and LATER.

What to look for in the report

Field	How to interpret it
Analysis / Engine / Run	The analysis mode, engine used and run identifier. Keep the Run ID to find the same result later.
DOCUMENTED FINDINGS	Findings supported by the records, each with a Source URL and a Locator showing where to look in the source.
Missing context topics	Missing topics such as access, exposure or weather. A complete topic checklist does not establish sufficient evidence.
RESEARCH NEXT STEPS / LIMITATIONS	Follow-up work and limits that must accompany your interpretation.

Agent trace shows the stages recorded for the selected run. “10/10” means all ten runs completed; it is not a measure of AI accuracy.

Export results and record a review

Save one case as a ZIP

1. Select the case at the top.
2. Click **Export case ZIP**, also at the top.
3. Choose a folder and save the file.
4. Extract the ZIP and open its **reports** folder.
5. Open the **.txt** file whose name matches the Run ID you want.

The export includes the saved runs for that case, not just the latest run. For Gironde, it also includes the available original pilot PDFs.

Save all ten cases as a ZIP

After the batch finishes, open **Analysis → Export all cases**. Save **FireWatch_All_10_Cases.zip**. Extract it to find one ZIP per case. Extract the case ZIP you want and open **reports/*.txt**.

Exporting does not run an analysis. Run the batch before exporting to include the new reports for all ten cases.

Inside each case ZIP	Contents
reports/*.txt	Reports readable in a text editor.
research_evidence.json / sources.csv	Prefilled documentary records and source catalogue.
runs.json / reviews.json / trace.jsonl	Runs, recorded reviews and stage history.
evidence.json	Manually added original records. It may be empty even when documentary or pilot evidence exists in other files.
pilot/ and originals/ (Gironde)	Pilot data and technical review, original maps and credits.

Record a human review

In **History & review**, select a run under **Saved run**. Enter your name in **Reviewer** and your justification in **Rationale**. Choose **REQUEST_CORRECTION** to request changes or **ACCEPT_RESEARCH_RECORD** to accept the research record. Click **Save research review**.

To accept the pilot, inspect the PDFs and check all three verification items only after completing those checks. The original report still displays PENDING; your decision is linked to the run and appears in the reviews, including reviews.json. Acceptance does not establish operational validation.

Gemini and troubleshooting

Enable Gemini after the offline test

1. Open **Gemini settings** and paste your key into **API key**.
2. Check the **Model ID**. The application defaults to the editable value gemini-2.5-flash; use a compatible model available to your account.
3. Click **Test Gemini connection**. A result of **source: gemini / reason: OK** means the program accepted the test response.
4. Check **Enable Gemini for analysis**.
5. Run the pilot or documentary analysis again. Opening an existing report does not add Gemini to that run.

Entering a key alone does not enable data transmission. When enabled, the program sends the admitted evidence text to Google's service. The connection test sends no case data. An authenticated API call must be verified on your computer.

In the report, **llm_assisted** means both stages returned accepted responses; **llm_partial_fallback** means one stage did; **deterministic_fallback** means the deterministic alternative was used. Stage results appear under GEMINI EVIDENCE and GEMINI CRITIC. Accepted responses still require human review.

If something looks wrong

Situation	What to do
A tab or button is hidden.	Maximize the window. Use Open view at the bottom to switch tabs.
I opened history but see no report.	Choose a run under Saved run, then open Analysis.
The original PDF will not open.	Check that originals was fully extracted and a PDF reader is available. If there is an integrity error, extract the complete original package again.
EARLY admits no evidence.	This is expected for that cutoff. Compare it with PRIMARY and LATER.
Historical cutoff requests data or blocks analysis.	For the ten cases, select Retrospective documents (prefilled). Historical cutoff requires original records and verified timestamps; the 30 records are not substitutes.
Gemini is slow or fails.	Wait for the bounded retries. Check the returned reason. To test local data, uncheck Enable Gemini and run again.
Launch or save error.	Use Python 3.12 with Tkinter, run the complete project folder and keep it in a writable location. Check the PyCharm Run output.